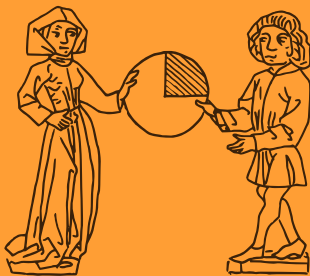
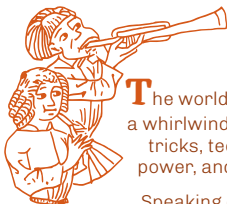


How to
S O U N D
S M A R T
When Talking About
CHARTS



DUNCAN BRADLEY



The world of charts is a whirlwind of debates, tricks, techniques, power, and potential.

Speaking eloquently about these challenges and opportunities isn't easy.

This handbook is intended as a guide to saying the right thing.

Here, the secret to success is letting others do the hard work for you. Memorise the words of clever people. Repeat their remarks at any opportunity.

Alternatively, these quotations may simply offer a peek into the mindset of data visionaries.

Let's begin...

“Graphs are like jokes: if you have to explain them, they have failed.”

— Sally Bigwood



A chart often requires lots of explanation to be understandable.

But the set-up needs to happen *before* the punchline.

Otherwise, the chart falls flat. The audience is left in stunned silence wondering what it was all about.

Designing effective graphics involves laying the groundwork for understanding: pre-empting sources of confusion, addressing concerns, providing the right level of detail.

A great chart probably won't get a laugh. But it might generate a subtle smile of appreciation.

Some people argue a graphic can't perfectly represent an entire table of numbers. And they're right. But that criticism misunderstands what charts offer.

It's precisely *because* charts condense the fine detail, and highlight the signal in the noise, that they make patterns memorable.

Recalling values to three decimal places is rarely useful. So, it's worth sacrificing some precision for the opportunity to digest the whole story. Through charts, datasets become *portable*.

**“To remember
simplified pictures is
better than to forget
accurate figures.”**

— Otto Neurath



See also: “You have to be like the worst tabloid newspaper in the front and the Academy of Science in the back.” — Hans Rosling

Simply plotting the values, and doing nothing else, might sound responsible. But this leaves interpretation to chance.

Instead, context is what gives numbers meaning. Without it, audiences won't find the patterns as obvious as expected.

So, an author must select the context *they* consider relevant. And that's where the idea of a neutral display disappears. Choosing nothing is still a choice: the chart will carry a message regardless.

In charts, it's important to lend one's voice to the data — and it's also unavoidable.

“The numbers have no way of speaking for themselves. We speak for them.”

— Nate Silver



See also: “We need to think much less about trust... much more about being trustworthy”

— Onora O'Neill

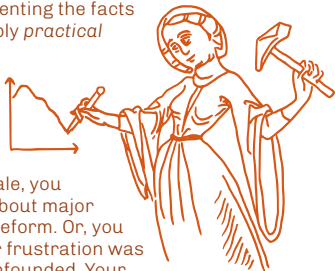
**“Whenever I am infuriated,
I revenge myself with a
diagram.”**

— Florence Nightingale

“Don’t let the sun go down on your anger” says the Bible. Easier said than done. Nightingale presents an unusual solution for rage: making charts.

Sit down, channel that negative energy into a nice little graphic.

All in all, presenting the facts is a remarkably *practical* response to injustice.



Like Nightingale, you might bring about major institutional reform. Or, you may find your frustration was completely unfounded. Your mileage may vary.

See also: *“Diagrams are a girl’s best friend.”* — Adam Dant



**“Fully explore
your dataset.
Then explore
it some more.”**

— Jen Christiansen

Captivating data graphics aren't just products of clever design.

Many of the creators of history's most famous charts weren't known for being great chart-makers. They were experts in their field of study.

Being immersed in a subject reveals the genuinely interesting, significant, surprising stories. And it provides the context required to tell them accurately.

If you don't happen to be an expert in the first place, endeavour to become one. Spend time with the statistics. Get to know the numbers.

See also: *“There is no method where you shake a dataset and insights fall out.”* — Gurman Bhatia

“Clarify, don’t simplify.”

— Nigel Holmes

See also: “*Visualisation is not something that happens on a page or on a screen; it happens in the mind.*” — Alberto Cairo

In a thesaurus, *clarify* and *simplify* might be considered synonyms. Yet, there’s a subtle distinction. Simple is about presentation, clear is about interpretation.



Charts don’t have to be sparse and austere to facilitate understanding. Clarity equips a reader with confidence, regardless of complexity.

Or, put another way: over-simplification is widespread — but it’s less common to find someone guilty of over-clarification.

“Data visualisation is actually about words.”

— John Burn-Murdoch

Why should text be an important aspect of making charts? Isn't a picture worth a thousand words?

Well, graphics on their own can be ambiguous — full of abstract forms without a clear meaning.

Text provides a helping hand by introducing, pointing out, reminding, clarifying, summarising.

Words are useful for conveying both the micro and the macro of a chart: the individual elements, and the core idea.

A little label beside a bar or line avoids uncertainty about what it refers to. And putting the main takeaway in writing reinforces the message, which also helps less-confident readers know what to look for when interpreting the graphic.

If in doubt, write it twice!



Bar charts are simple, familiar, precise. All good qualities. But what about eye-catching, memorable, emotive?



Depicting data isn't all about efficiency. There's also value in breaking the mould.

The alternative is crafting a graphic that lends itself to a different type of interpretation. One that's perfectly tailored to the numbers and the subject matter. A design that's a pleasure to look at.

Let's live in a world with charts that make you stop and look. Let there be joy!

“There’s a strand of the data viz world that argues that everything could be a bar chart. That’s possibly true but also possibly a world without joy.”

— Amanda Cox

**“Grey is
the most
important
colour in data
visualisation.”**



— Lisa Charlotte Muth

Want a pattern to stand out? Use grey.

Colour is excellent at attracting attention. But if there's too much, everything shouts, and nothing is heard.

Grey is perfect for everything with a supporting role: axes, gridlines, labels, supplementary text.

It's also fantastic for showing context: the other bits of data that contribute to the story, build understanding, put things in perspective.

Saturated hues establish the key players. Desaturated hues help keep the extras out of the spotlight.

See also: *“Admit colours into charts gracefully, as you would receive in-laws into your home.”* — Dona M. Wong

A chart-maker wears many hats: designer, statistician, psychologist. In the modern age, they often write code, too.

The work is therefore a delicate balancing act: multiple perspectives, competing priorities.

On the software side, every nuance ought to be considered. The code must reflect the complexity in the dataset and the design. Ignored edge-cases cause problems.



Details matter for designers, too, of course. But obsessing over pixels is a dangerous game. Endless tweaks mean infinite delays. Keeping sight of the bigger picture is an essential skill.

The trick — like in fashion — is to wear the right hat at the right time.

“Bad programmers ignore details. Bad designers get lost in details.”

— Nate Kirby

One word has come to dominate conversations about the purpose of charts: insight.

“Insight is overrated.”

Certainly, charts can inspire eureka moments — but they do lots more, too.

— RJ Andrews

They can also remind someone what they already know or simply show what's being measured. They can demonstrate trustworthiness through transparency, capture attention, provide a sense of scale, and much more.

This creates opportunities to put charts to use more effectively; to take advantage of *everything else* they're capable of.

A chart might be completely forgotten once it's started a conversation. But that doesn't mean it wasn't valuable.



Charts are a unique form of communication. They are tools for interpreting the evidence oneself, supporting independent enquiry.

People with visual and cognitive impairments deserve that experience too.

Making *inclusive* charts involves being flexible about methods of presentation. That means adjusting a design to accommodate varying abilities, and offering alternatives which present the same account of the data.

Yes, that requires effort — because it's part of the work.



**“Accessibility
isn’t extra
steps, it’s
steps you’ve
missed.”**

— Nick Tillem

See also: “You probably can’t write text that conveys the entire meaning of a chart. But, that doesn’t mean that you shouldn’t try.” — Amy Cesal

There's a certain absurdity in turning numbers into pictures. Typically, it involves rendering an aspect of our rich and complex world using straight lines and abstract shapes.

That isn't the only option.

Depicting subjects in a more *concrete* way can remind readers of their importance: there are real experiences behind the statistics.

Treat data with the respect it deserves. If it's about people, consider making those people visible. If it's about farts...

**“If it's
about farts,
draw a butt
for God's
sake.”**

— Mona Chalabi



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